Chapter 3 solutions

CH03-E1

1. S_1 is a line through the origin. For $(x, 2x)^T, (z, 2z)^T$ and $c \in \mathbb{R}$,

$$(x, 2x)^T + (z, 2z)^T = (x + z, 2x + 2z)^T \in S_1,$$
$$c(x, 2x)^T = (cx, 2cx)^T \in S_1.$$

Also $0 \in S_1$, so it is a subspace.

2. S_2 is not a subspace. Both $(1, 0)^T$ and $(0, 1)^T$ belong to it, but

$$(1, 0)^T + (0, 1)^T = (1, 1)^T \notin S_2.$$

Addition is not closed. Containing zero alone is insufficient.

3. S_3 is not a subspace. It contains $(1, 0)^T$, but multiplication by 2 gives $(2, 0)^T \notin S_3$. It also omits zero. The unit circle fails the linear closure conditions.

CH03-E2

1. Subtract twice row 1 from row 2, and row 1 from row 3.

$$A \to \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

No further pivot-column entries need clearing, so RREF is

$$R = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The rank is 2. Pivot columns are 1 and 3; free columns are 2 and 4.

2. Set $x_2 = s, x_4 = t$. Then

$$x_1 + 2s + t = 0, \qquad x_3 + t = 0.$$

Therefore

$$x = (-2s - t, s, -t, t)^T = s(-2, 1, 0, 0)^T + t(-1, 0, -1, 1)^T.$$

A basis is

$$\mathcal{B}_{N(A)} = \{(-2, 1, 0, 0)^T, (-1, 0, -1, 1)^T\}.$$

3. Direct multiplication gives

$$A(-2, 1, 0, 0)^T = (-2 + 2, -4 + 4, -2 + 2)^T = 0,$$
$$A(-1, 0, -1, 1)^T = (-1 + 1, -2 - 1 + 3, -1 - 1 + 2)^T = 0.$$

If $c_1 s_1 + c_2 s_2 = 0$, coordinate 2 forces $c_1 = 0$ and coordinate 4 forces $c_2 = 0$. Hence the vectors are independent.

CH03-E3

Augment the matrix by b.

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & 1 & 3 & 5 \\ 1 & 2 & 1 & 2 & 3 \end{array}\right].$$

Subtract twice row 1 from row 2, and row 1 from row 3.

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right].$$

Subtract row 2 from row 3. The left block is RREF and the right block is $(2, 1, 0)^T$. There is no contradictory row, so the system is consistent.

Set $x_2 = x_4 = 0$. Then

$$x_1 = 2, \qquad x_3 = 1$$

giving $x_p = (2, 0, 1, 0)^T$.

The complete solution is

$$x = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{R}.$$

Since $b \neq 0$, zero is absent. This is a translate of $N(A)$, not a subspace.

CH03-E4

1. Original columns 1 and 3 are the pivot columns.

$$\mathcal{B}_{C(A)} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

They are independent, so they form a basis and $\dim C(A) = 2$.

2. The nonzero RREF rows give

$$\mathcal{B}_{C(A^T)} = \{(1, 2, 0, 1)^T, (0, 0, 1, 1)^T\}.$$

At coordinates 1 and 3, these vectors have patterns $(1, 0)$ and $(0, 1)$. They are independent, so $\dim C(A^T) = 2$.

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CH03-E4 (continued)

3. Solve $A^T y = 0$ with $y = (a, b, c)^T$. The first three equations are below. The fourth, $a + 3b + 2c = 0$, is the sum of the first and third and adds no constraint.

$$a + 2b + c = 0, \quad 2a + 4b + 2c = 0, \quad b + c = 0.$$

The second equation repeats twice the first. The third gives $c = -b$, and the first then gives $a = -b$. Hence

$$N(A^T) = \text{span}\{(-1, 1, -1)^T\},$$

This gives a one-vector basis.

4. Since $m = 3, n = 4, r = 2$,

$$r + (n - r) = 2 + 2 = 4 = n,$$
$$r + (m - r) = 2 + 1 = 3 = m.$$

The first sum pairs row space and nullspace in $\mathbb{R}^4$. The second pairs column space and left nullspace in $\mathbb{R}^3$.

CH03-E5

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The trigonometric identity gives

$$\sin^2 x = \frac{1 - \cos(2x)}{2}.$$

Therefore

$$\frac{1}{2}v_1 - \frac{1}{2}v_2 - v_3 = 0$$

for every x. The coefficients $(1/2, -1/2, -1)$ are not all zero, proving dependence.

Since v_3 is generated by v_1, v_2, the span has dimension at most 2. If $a + b\cos(2x) = 0$ for every x, setting $x = 0$ gives $a + b = 0$, and setting $x = \pi/4$ gives $a = 0$, hence $b = 0$. Thus v_1, v_2 are independent and form a basis of a two-dimensional span.

CH03-E6

1. With $m = 3, n = 5, r = 2$, the dimensions are as follows.
   - $C(A) \subseteq \mathbb{R}^3$, dimension 2;
   - $C(A^T) \subseteq \mathbb{R}^5$, dimension 2;
   - $N(A) \subseteq \mathbb{R}^5$, dimension $5 - 2 = 3$;
   - $N(A^T) \subseteq \mathbb{R}^3$, dimension $3 - 2 = 1$.
2. Every consistent solution set is $x_p + N(A)$. Since $\dim N(A) = 3$, it has three free parameters.
3. Yes. Rank $2 < 3$ makes $C(A)$ a proper plane in $\mathbb{R}^3$. Every b outside it is inconsistent. Nonzero homogeneous solutions do not imply that all right-hand sides are attainable.

CH03-E6 (continued)

4. No. If $\dim N(A) = 0$, rank-nullity would give

$$r = 5 - \dim N(A) = 5.$$

This exceeds the number of rows, 3, and contradicts the given rank 2. In fact nullity equals 3.

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CH03-E7

Let $c = (c_1, c_2, c_3)^T$. Independence of the given basis means that

$$c_1 w_1 + c_2 w_2 + c_3 w_3 = 0$$

is equivalent to $Bc = 0$.

1. B has pivots in columns 1 and 2 and rank 2. Thus $Bc = 0$ has a nonzero solution, so the three w_j are dependent.
2. The coordinate isomorphism preserves span dimension, giving dimension $\operatorname{rank} B = 2$. Here $w_1 = u_1, w_2 = u_2$. Their coordinate vectors are the first two standard basis vectors, so they form a basis of this span.
3. The third column of B is the sum of its first two columns.

$$\operatorname{col}_3(B) = \operatorname{col}_1(B) + \operatorname{col}_2(B).$$

Hence $B(1, 1, -1)^T = 0$, giving

$$w_1 + w_2 - w_3 = 0.$$

Substituting $w_3 = w_1 + w_2$ verifies the relation directly. The coefficients $(1, 1, -1)$ are nonzero as a vector.

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CH03-E8

1. Using the four basis vectors from the chapter,

$$u_1 = (1, 2, 0, 1)^T, \quad u_2 = (0, 0, 1, 1)^T,$$
$$s_1 = (-2, 1, 0, 0)^T, \quad s_2 = (-1, 0, -1, 1)^T.$$

Write $x = au_1 + bu_2 + cs_1 + ds_2$. Equating coordinates yields $a - 2c - d = -2$, $2a + c = 3$, $b - d = 0$, and $a + b + d = 3$. Elimination gives $b = (3 - a)/2$ and $5a - b = 4$, hence $a = b = c = d = 1$. Therefore

$$x_{\text{row}} = u_1 + u_2 = (1, 2, 1, 2)^T,$$
$$x_{\text{null}} = s_1 + s_2 = (-3, 1, -1, 1)^T.$$

So $x = (-2, 3, 0, 3)^T$.

2. x_{row} is the sum of the two nonzero RREF rows, hence belongs to $C(A^T)$. Also

$$As_1 = As_2 = 0,$$

so $x_{\text{null}} \in N(A)$. Their sum is x.

CH03-E8 (continued)

3. Suppose

$$\alpha u_1 + \beta u_2 + \gamma s_1 + \delta s_2 = 0,$$

Comparing coordinates gives

$$\gamma = -2\alpha, \quad \delta = \beta, \quad 5\alpha - \beta = 0, \quad \alpha + 2\beta = 0.$$

Thus $\beta = 5\alpha$; substitution in the fourth equation gives $11\alpha = 0$, so all coefficients vanish. These four independent vectors form a basis of $\mathbb{R}^4$, giving a decomposition for every x. The difference of two decompositions is a zero combination of this basis, forcing every coefficient difference to vanish and proving uniqueness.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 3.

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